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CLINICAL REVIEW
Novel therapeutic concepts

Immunotherapy for cardiovascular disease

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The outcomes of the Canakinumab Anti-inflammatory Thrombosis Outcome Study (CANTOS) trial have unequivocally proven that inflammation is a key driver of atherosclerosis and that targeting inflammation, in this case by using an anti-interleukin-1β antibody, improves cardiovascular disease (CVD) outcomes. This is especially true for CVD patients with a pro-inflammatory constitution. Although CANTOS has epitomized the importance of targeting inflammation in atherosclerosis, treatment with canakinumab did not improve CVD mortality, and caused an increase in infections. Therefore, the identification of novel drug targets and development of novel therapeutics that block atherosclerosis-specific inflammatory pathways and exhibit limited immune-suppressive side effects, as pursued in our collaborative research centre, are required to optimize immunotherapy for CVD. In this review, we will highlight the potential of novel immunotherapeutic targets that are currently considered to become a future treatment for CVD.

Keywords Cardiovascular disease • Coronary artery disease • Inflammation • Novel therapies • Novel targets • Cytokines

The immune system and atherosclerosis

Atherosclerosis, the underlying cause of the majority of cardiovascular diseases (CVDs), is a lipid-driven, inflammatory disease of the large arteries.[1] Although our insights in lipid metabolism and the development of lipid-lowering drugs, in particular HMG-CoA reductase inhibitors (statins) and the more recently developed proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors has helped to lower the incidence of CVDs substantial part of the population still suffers from CVD notwithstanding optimal lipid-modulating therapy.[2] Beyond yet to be identified mechanisms, e.g. involving variation of genetic or epigenetic factors, this indicates that inflammation, the other main driver of atherosclerosis, plays a major role in a subset of CVD patients. Indeed, the interactions of the different innate and adaptive immune cells, and the subsequent secretion of immune-regulatory and activating cytokines and chemokines have been found to determine the progression of atherosclerosis in experimental models.[3–8] The interplay between lipids, lipid-lowering therapy and inflammation has been discussed in a recent ESC position paper, concluding that inflammation is an important regulator of atherosclerosis, both dependent and independent of lipids.[9] Another position paper elaborates that inflammatory markers, especially high-sensitivity C-reactive protein (hs-CRP), are among the most valid biomarkers for atherosclerotic CVD.[10]

Recently, the Canakinumab Anti-inflammatory Thrombosis Outcome Study (CANTOS) trial has proven that targeting a pro-inflammatory mediator, interleukin (IL)-1β, effectively reduces CVD risk and mortality in patients with residual inflammation, without altering low-density lipoprotein (LDL) concentrations.[11] In contrast, treating CVD with methotrexate, a broad-spectrum anti-inflammatory agent, failed to reduce CVD risk or mortality.[12] Therefore, the identification of novel drug targets that block atherosclerosis-specific inflammatory pathways is of utmost importance. In this review, we will discuss the current status on the potential of novel immunotherapeutic targets for atherosclerosis.

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Immunotherapy for atherosclerosis and cardiovascular disease: the IL-1β-IL6-CRP axis

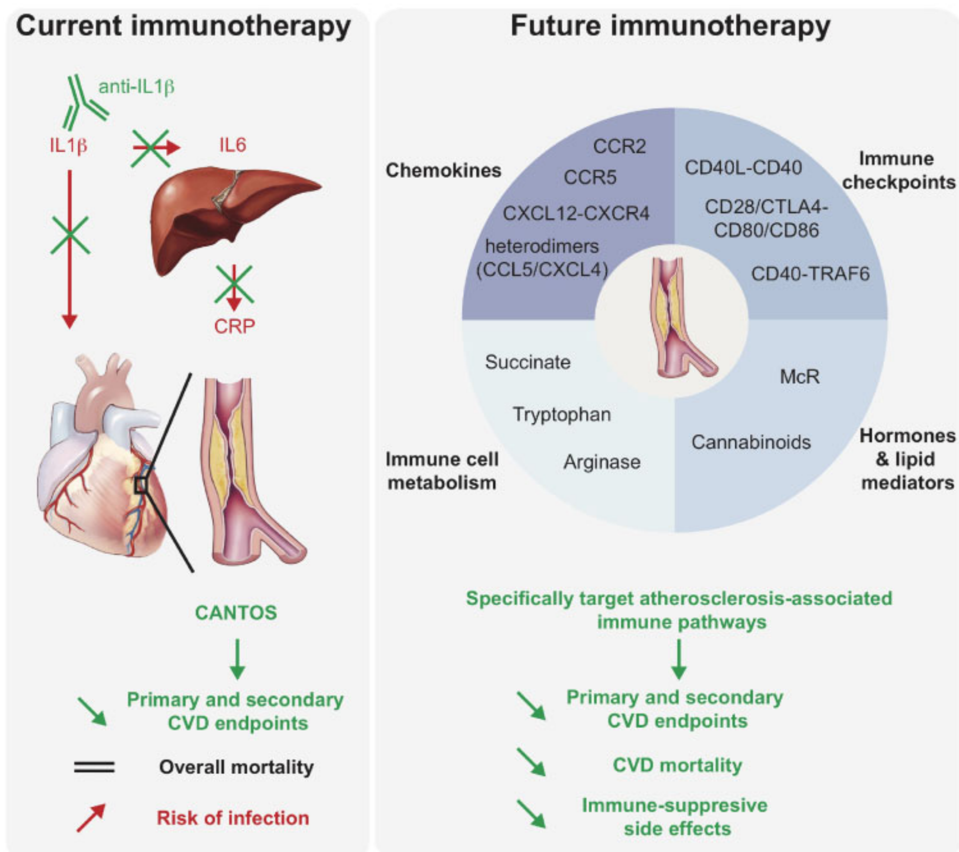
Cytokines are important mediators of inflammation that can activate or modulate immune reactivity. The IL-1 family of proteins, which includes the pro-inflammatory IL-1α and IL-1β isoforms, as well as the anti-inflammatory IL-1 receptor antagonist (IL1ra), mediate atherosclerosis via multiple actions, including activation of endothelial cells, vascular smooth muscle cells (VSMCs) and innate immune cells.[13] In atherosclerotic mice, IL-1 generally promotes lesion formation, and interruption of IL-1 signalling limits atherogenesis. Especially the IL-1β isoform is a powerful mediator of atherosclerosis aggravating processes. Interleukin-1β is activated via the NLP Family Pyrin 3 (NLRP3) inflammasome, a multi-protein assembly that integrates inflammatory signals provoked by cholesterol crystals, hypoxia, and turbulent flow.[13] Inhibition of the NLRP3 inflammasome in mice results in a strong reduction in atherosclerosis.[14] Once activated, IL-1β stimulates vascular endothelial cells to express cell surface proteins that increase inflammatory cell adhesion, increases VSMC proliferation and macrophage activation. Interleukin-1β also up-regulates IL-6, another pro-inflammatory cytokine that induces hepatocytes to synthesize and release different acute phase reactants, including C-reactive protein (CRP), fibrinogen, and plasminogen activator inhibitor.[13] Thus, IL-1β lies upstream in the inflammation pathway and thereby modulates numerous inflammatory components of atherosclerosis (*Take home figure*).

The CANTOS trial provided convincing evidence that targeting IL-1β effectively reduces CVD risk and mortality in patients with residual inflammation as indicated by hs-CRP >2 mg/dL, although only at one dose and not in terms of single endpoints. The extent of risk reduction for major cardiovascular events, cardiovascular death, and death from any cause with canakinumab was greatest among patients with the largest reductions in levels of IL-6 and hs-CRP,[15] which suggests that the benefit was related to the targeting of the IL-1β–IL-6–CRP pathway of innate immunity.[16] These effects were independent from changes in LDL concentrations. Notably, this came at the expense of adverse effects, namely a slight, but significant increase in the number of fatal infections.[11] The results of CANTOS thus deliver proof-of-principle that inhibiting inflammation can prevent atherosclerosis-related events in humans.[17] Prompting caution for use in late-stage atherosclerosis, recent results in mice reveal that IL-1β exerts multiple protective effects in advanced atherosclerosis, promoting outwards remodelling and a stable fibrous cap.[18] On the contrary, the Cardiovascular Inflammation Reduction Trial (CIRT), which tested low-dose methotrexate in patients with a history of myocardial infarction (MI) and type 2 diabetes or metabolic syndrome failed to reduce CVD events and to lower CRP, IL-1β, and IL-6 levels, suggesting that therapies for atherosclerosis should be aimed at specific and atherosclerosis-associated inflammatory pathways.[12] Emerging evidence suggests beneficial cardiovascular effects of anti-inflammatory agents in patients with rheumatoid arthritis (RA). *Table 1* summarizes phase 3 and approved cardiovascular outcome studies with anti-inflammatory agents in patients with atherosclerosis and/or rheumatoid arthritis. The placebo-controlled LoDoCo study introduced the use of colchicine in secondary prevention of coronary artery disease (CAD), randomizing patients with stable CAD to colchicine vs. control to reveal a significant reduction in recurrent cardiovascular events over a 3-year follow-up period.[19] While two more clinical trials are currently scrutinizing the efficacy of colchicine in CVD (COLCOT, NCT02551094 and LoDoCo2, ACRTN12614000093684), drugs specifically targeting causally relevant drivers of atherosclerosis that may reduce CVD risk even further than canakinumab are eagerly awaited. This is particularly true after the decision of Novartis to not pursue canakinumab in CVD (but rather anti-inflammatory treatment in cancer), which likely reflects the unfavourable ratio between the limited therapeutic window at high cost and adverse effects in CVD, as this may be further aggravated when lipid therapy with PCSK9 siRNA becomes available. Yet, even under optimal lipid lowering, the subgroup benefiting from anti-inflammatory treatment calls for cost-effective therapeutics without immune-related side effects.

It is important to note that immunotherapy in CVD has experienced numerous limitations in various other areas. For instance, blocking the terminal complement component with pexelizumab failed to affect outcomes in patients with acute MI.[20] Neither did the preliminary phase of a large-scale clinical trial with the MAPK inhibitor losmapimod (LATITUDE) yield clinical benefits, prompting discontinuation of this target.[21] Inflammation also plays an important role in cardiac remodelling and thereby in the pathogenesis of heart failure (HF),[22] but until recently, clinical studies have failed to improve HF (*Table 2*). Despite an increase of tumour necrosis factor-alpha (TNF-α) levels in advanced HF, the treatment with anti-TNF-α has been shown to be ineffective or even worsen the prognosis of HF in randomized controlled trials (ATTACH, RENNAISSANCE, and RECOVER).[23,24] Likewise, the Celacade device claiming to unspecifically boost anti-inflammatory immunomodulation in patients with chronic HF did not meet primary endpoints in the ACCLAIM trial.[25] Interestingly, a sub-study from the CANTOS trial showed that anti-IL-1β treatment reduced the number of post-MI patients with hs-CRP levels that were hospitalized for HF, and also reduced HF-related mortality in this patient group, indicating that immunotherapy might be beneficial for a selected group of HF patients (*Table 2*).[26] Recent experimental data revealed that, besides the innate immune system, adaptive immunity also plays a crucial role in HF. Effector T cells are detrimental for proper cardiac remodelling after MI and suppression of effector T-cell responses via regulatory T cells has proven to be beneficial in experimental HF.[27] Thus, further research exploring the potential of immunotherapy for HF is clearly warranted. Beyond these exemplary caveats and considerations, we will also not cover the arduous attempts at protective vaccination against atherosclerosis in this review.

Immunotherapy for atherosclerosis and cardiovascular disease: the next generation of targets

The outcomes of CANTOS and CIRT have taught us a wise lesson: immunotherapy for CVD can only be applied when specific



Take home figure Immunotherapy for cardiovascular disease. Anti-IL-1β antibody (canakinumab) treatment is currently the only immunotherapy available to treat cardiovascular disease. Interleukin-1β is a powerful driver of atherosclerosis. Once activated, IL-1β activates not only macrophages, but also endothelial cells and vascular smooth muscle cells. It also up-regulates IL-6, another pro-inflammatory cytokine that induces hepatocytes to synthesize and release different acute phase reactants, including C-reactive protein. Anti-IL-1β therapy improved primary and secondary cardiovascular disease outcomes but failed to improve overall mortality and caused an increased risk of infection. Therefore, novel powerful immunotherapeutics are being developed. Potent targets are chemokines and their receptors, immune checkpoints, immune cell metabolism, and hormones and lipid mediators. These novel immunotherapeutics are expected to not only improve primary and secondary cardiovascular disease endpoints, but also cardiovascular disease mortality without immune-suppressive side effects.

inflammatory pathways that are crucial for the pathogenesis for atherosclerosis are targeted.[11,12] The inflammatory process in atherosclerosis features the activation of a network of immune cells and inflammatory pathways. Research during the past three decades has revealed that besides the IL1β-IL6-CRP axis, four other inflammatory pathways are crucial in atherosclerosis and have the potential to become a therapeutic drug target for the treatment of CVD in the near future. These pathways include *chemokines and their receptors*, *immune checkpoints*, *pathways of immune-metabolism*, and *hormones and lipid mediators* (*Take home figure*).

Chemokines and their receptors

A cytokine-related class of immune regulators are the small *chemotactic cytokines* (8–12 kDa), also known as chemokines. Chemokines are key regulators of (immune) cell traffic in a steady state and during inflammation and are classified according to their conserved cysteine residues which correspond to their categorized G protein–coupled receptors. Upon ligand binding these receptors activate downstream G protein-coupled signalling, while newly categorized atypical chemokine receptors are able to internalize, scavenge, or transport chemokines.[28] This promiscuous and complex system comprises approximately 50 ligands and 20 receptors expressed in mammals, which orchestrate migration and different immune functions of many cell types crucial in atherosclerosis and CVD. Thus, blocking of specific chemokine-receptor dyads holds promising therapeutic targets (*Figure 1A*).[1]

Prior to atherosclerotic plaque formation modified lipids (e.g. oxidized LDL) trigger arterial endothelial dysfunction, for example through LDL-components such as lysophosphatidic acids which induce CXCL1 release from endothelial cells. CXCL1 in turn interacts with CXCR2 on myeloid cells thereby stimulating their mobilization to the blood and sites of chronic inflammation, especially under hyperlipidaemic conditions. Accordingly, haematopoietic CXCR2-

Table 1 Clinical trials and inception studies involving anti-inflammatory therapies for CVD (A) and rheumatoid arthritis (B) with beneficial effects on cardiovascular outcome

Therapeutic/study name	Agent	Method of action	Sponsor	Outcome	Status/references
(A) Anti-inflammatory therapies for CVD					
CANTOS (NCT01327846)	Canakinumab	IL-1β antibody	Novartis	Reduction in recurrent CV events	Completed Ridker et al.[11]
CIRT (NCT02576067)	Methotrexate	Purine metabolism inhibitor	Icahn School of Medicine at Mount Sinai	No benefit	Completed Ridker et al.[12]
COLCOT (NCT02551094)	Colchicine	Tubulin disruption	Montreal Heart Institute	Ongoing	Phase III
Inflammation and vascular function in atherosclerosis (NCT00760019)	Salsalate	IκB kinase inhibitor	Brigham and Women's Hospital	Impaired flow-mediated endothelium dependent vasodilation	Completed Nohria et al.[117]
Immune modulating therapies					
ABC (NCT00539266)	Autologous transplantation	Bone marrow-derived mononuclear cells	Leiden University Medical Center	No benefit	Completed
LDL immunization					
V6 (NCT03042741)	V6	Vaccination	Immunitor LLC	Ongoing	Phase III
(B) Anti-inflammatory therapies for rheumatoid arthritis					
Protocare longitudinal health benefit claims database (Aventis)		DMARDs	Hospitalization for CHF	Reduction in hospitalization for CHF	Bernatsky et al.[118]
CORRONA registry (Abbott, Amgen, BMS, Centocor, Genentech, Lilly and Roche)		TNF-α antagonist	All CV events MI	Reduced risk of all CV events	Greenberg et al.[119]
Regional registry	etanercept or infliximab	TNF-α antagonist	All CV events	Reduced risk of all CV events	Jacobsson et al.[120]
Meta-analyses: MEDLINE, EMBASE, Cochrane		TNF-α antagonist	All CV events MI Stroke MACE	Reduced risk of all CV events in RA	Roubille et al.[121]
Meta-analyses: MEDLINE, EMBASE, Cochrane	Methotrexate	Purine metabolism inhibitor	All CV events MI	Reduced risk of all CV events in RA	Roubille et al.[121]

CHF, congestive heart failure; CV, cardiovascular; CVD, cardiovascular disease; DMARDs, disease-modifying anti-rheumatic drugs; LDL, low-density lipoprotein; MACE, major adverse cardiac events; MI, myocardial infarction; RA, rheumatoid arthritis; TNF-α, tumour necrosis factor-alpha.

deficiency as well as systemic absence of CXCL1 protect from atherosclerosis and reduce plaque macrophage content in mice.[29] The mobilization of classical/inflammatory monocytes is tightly regulated by the CCL2/CCR2 axis and CCR2-deficient mice were among the first models reported to develop less atherosclerotic lesions. Simultaneous deletion of CCL2 and CX3CR1, or CCR2 and CX3CL1 further decreased lesion burden most likely due to an stronger attenuation of monocytosis, resulting in lower plaque macrophage numbers.[1,30] In accordance siRNA-mediated silencing of CCR2 or administration of a non-agonistic CCL2-competing mutant (PA508) with increased proteoglycan affinity into mouse models of MI reduced classical monocyte recruitment to the infarcted area and suppressed left ventricular remodeling[31] or attenuated ischaemia/reperfusion injury in mice.[32] In humans, blocking of CCR2 with MLN1202, a highly specific humanized monoclonal antibody that inhibits CCL2 binding, reduced CRP levels in patients at risk for CVD in a phase 2 trial and remains a promising target of future therapies.[33]

Interaction of chemokines and their receptors does not only trigger leucocyte recruitment but also integrin activation and leucocyte arrest on the endothelium, particularly through CCL5 and CXCL4. Activated platelets deposit CCL5 on inflamed arterial endothelium, resulting in leucocyte recruitment.[30] Binding of CCL5 to CCR1 or CCR5 requires receptor sialylation for improved ligand-receptor interaction, and mice with a deficiency in sialyltransferase St3Gal-IV indeed displayed reduced arterial monocyte adhesion and atherosclerotic lesion sizes in a CCL5-related manner.[34] Consequently, CCR5 deficiency or interference with CCR5 by applying Met-CCL5 or inhibiting CCR5 with the FDA-approved HIV-entry inhibitors

Table 2 Clinical trials involving anti-inflammatory therapies for heart failure

Therapeutic/study name	Agent	Method of action	Sponsor	Outcome	Status/references
Anti-inflammatory therapies for heart failure					
ATTACH	Infliximab	Anti-TNF-α antibody	Centocor	Increased risk of death or hospitalization for CHF on high dose infliximab	Pilot, Chung *et al.*[23]
RENEWAL (RENAISSANCE + RECOVER)	Etanercept	Recombinant TNF receptor	Amgen, Wyeth	No benefit	Completed, Mann *et al.*[24]
ACCLAIM (NCT00111969)	Immunomodulation therapy	Oxidized autologous blood	Vasogen	Reduction in CV risk	Completed, Torre-Amione *et al.*[25]
REDHART (NTC01936909)	Anakinra	Recombinant IL1RA	NHLB	Improved peak VO_2 after 12 weeks treatment	Completed, van Tassel *et al.*[122]
CANTOS (NCT01327846)	Canakinumab	IL-1β antibody	Novartis	Reduction in hospitalization for HF and HF mortality	Completed, Everett *et al.*[123]

CV, cardiovascular; HF, heart failure; IL, interleukin; TNF-α, tumour necrosis factor-alpha.

maraviroc reduced plaque burden and plaque macrophage content in mice.[30] Clinically, maraviroc treatment of HIV-patients seems to lower atherosclerotic lesion growth[35] and a correlation between plasma CCL5 levels and progression of atherosclerosis in 204 patients after acute coronary syndrome or healthy controls has been reported.[36]

An interesting feature of chemokines in terms of drug development is their capacity to form higher-order oligomers with themselves (homomers) or other proteins (heteromers). In this context, CCL5 complexes with CXCL4 augmented CCL5-evoked arterial monocyte adhesion while selective disruption of CCL5–CXCL4 heterodimer formation with the cyclic peptide MKEY markedly reduced murine atherosclerotic lesion formation.[37] Application of MKEY has also been reported to preserve heart function after MI, to inhibit experimental aortic aneurysm initiation and progression and to protect against stroke in mice.[38] Another example is neutrophil-borne human neutrophil peptide 1 (HNP1) which forms heterodimers with platelet-derived CCL5 to stimulate monocyte adhesion through CCR5 ligation. In turn, disrupting HNP1-CCL5 interactions selectively with a specific peptide named SKY mitigated myeloid cell recruitment in a mouse model of MI.[39]

Another crucial chemokine-receptor axis in cell homeostasis, mobilization and immunity is the CXCR4-CXCL12 pair. Systemic treatment with the biglycan CXCR4 antagonist AMD3100 or reconstitution with CXCR4-deficient bone marrow aggravated diet-induced atherosclerosis by promotion of severe leucocytosis in mice. In contrast, microRNA-126 can promote autocrine endothelial CXCL12 expression driving CXCR4-dependent progenitor cell recruitment to limit atherosclerotic lesion formation in mice.[40] Applying cell-specific CXCR4 deletion in mice, endothelial CXCR4 was found to promote re-endothelialization after injury and to prevent neointimal hyperplasia.[31,32] Furthermore, CXCR4 in endothelial cells limits atherosclerosis by preserving endothelial barrier function, while CXCR4 in smooth muscle cells maintains a normal and contractile phenotype.[41] Regression analyses in humans identified the C-allele at rs2322864 within the *CXCR4* locus to be associated with increased risk for coronary heart disease, and to correlate with reduced CXCR4 expression in carotid artery plaques and a higher prevalence of symptomatic disease.[41] Notably, expression of CXCR4 and CXCL12 was increased in both stable and unstable carotid atherosclerotic plaques compared with healthy vessels, which was related to macrophage content.[42] Genome-wide association studies indeed established that single nucleotide polymorphisms at 10q11 near *CXCL12*, namely intergenic rs2802492, were independently associated with the risk of CAD and with CXCL12 plasma levels.[43,44] A Mendelian randomization study also confirmed CXCL12 as a causal mediator of CAD in the populations of ORIGIN and CARDIoGRAM,[45] whereas cell-specific deletion models identified that CXCL12 derived from endothelial cells drives atherosclerosis in mice.[43,44] However, while the CXCR4 antagonist Plerixafor (AMD3100) for haematopoietic stem cell mobilization was FDA-approved in 2008,[40] clinical trials probing effects of CXCR4/CXCL12 interference with respect to CVD have not been reported to date.

Chemokines and their corresponding ligands are an integral part of the immune system and have proven to be strong drivers and regulators of CVD. However, interference with this system to reduce CVD has to be context-specific and should be carefully designed. Promising approaches could be the blocking of multiple chemokines applying high-affinity antibodies,[46] or disruption of heterophilic interactions between disease-driving chemokines, the latter of which has already been proven successful in experimental atherosclerosis models.[37,47] Hence, tailored manipulation of heteromer formation may be the more promising approach, as such interventions can leave normal physiological and immunological functions intact.[37,47]

Immune checkpoints

Immune checkpoint proteins, including co-stimulatory and co-inhibitory proteins, are key regulators of inflammation and were

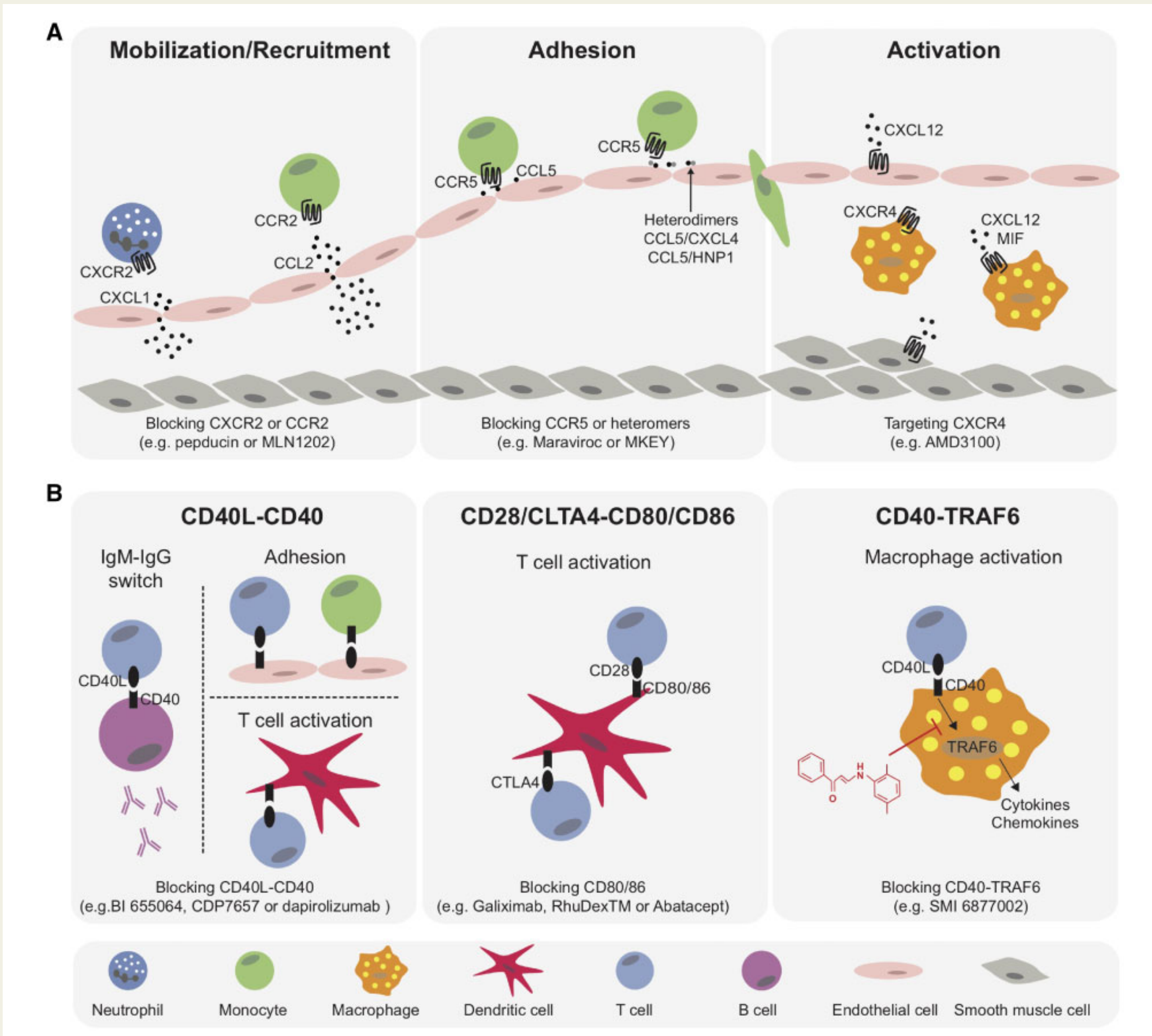


Figure 1 Targeting chemokine-receptor axes and immune checkpoints in atherosclerosis. (*A*) Activated endothelium expresses the chemokine CXCL1 causing myeloid cells bearing CXCR2 to arrest and migrate on the endothelium. Likewise, CCL2 triggers recruitment and adhesion of myeloid cells expressing CCR2. The chemokine receptor CXCR4 mediates migration of leucocytes towards its endothelial ligand CXCL12 or macrophage migration inhibitory factor. Interaction of CCR1 and CCR5 on leucocytes with CCL5 presented on endothelial cells mediates their adhesion and migration. This can be enhanced by heterodimer formation of CCL5 with CXCL4 or human neutrophil peptide 1. Potential targeting strategies for human disease are indicated. (*B*) CD40-CD40 ligand interactions drive atherosclerosis via different mechanisms, including T-cell activation via co-stimulation, leucocyte adhesion, and Ig-isotype switching. CD28–CD80/86 interactions mediate T-cell activation and antigen-presenting cell activation in atherosclerosis via co-stimulation. CTLA4-CD80/86 interactions counteract this activation via a co-inhibitory response. The interaction via CD40 and TRAF6 in macrophages drives macrophage activation and atherosclerosis. This interaction can be blocked by using small molecule inhibitor 6877002. Potential targeting strategies for human cardiovascular disease are indicated.

originally described to propagate or halt immune-cell activation when T cells and antigen-presenting cells interact (*Figure 1B*). Following recognition of a specific antigen via T-cell receptor (TCR), signalling via co-stimulatory molecules induces T-cell activation, which is then further augmented by chemokines and cytokines. Co-inhibitory receptor-ligand dyads dampen the activation of T cells and are required to maintain tolerance to self-antigens.[48,49] However, we now know that immune checkpoint proteins are present and functional on multiple innate and adaptive immune cells, as well as on endothelial cells, VSMCs, adipocytes, platelets, and epithelial cells.[50]

In the past decade, immune-therapy using immune checkpoint modulation, especially inhibition of programmed cell death protein (PD) 1 and cytotoxic T lymphocyte associated protein (CTLA) 4, has revolutionized the field of cancer therapy.[51] However, their exploitation in atherosclerosis is only just beginning. A plethora of immune checkpoints, including CD28, CD80/86, CTLA4, CD40, CD40L, Ox40L, Ox40, CD27, CD70, PD1, PDL1/2, inducible T-cell co-stimulator (ICOS), CD30 and CD137L is involved in atherogenesis, each via their unique mechanisms. The balance between the cell-type or subset-specific actions of these ligand-receptor pairs, e.g. effector or regulatory T cell, macrophage, dendritic cell (DC), or B cell within the network determines whether the immune checkpoint protein promotes or limits atherosclerosis.[49,50,52] Moreover, the role of regulators of the immune checkpoint landscape, such as the E3-ligase Casitas B-cell Lymphoma B (CBL-B) in CVD have only recently been explored.[53]

The CD80/86-CD28 and CD80/86-CTLA4 immune checkpoint proteins are pivotal regulators during atherosclerosis and drive or inhibit plaque inflammation. Human atherosclerotic plaques contain $CD80/86^{+}$ macrophages, and $CD28^{+}$ T cells, with higher expression levels of CD80 and CD86 in vulnerable compared with stable plaques.[54] Hyperlipidaemic mice deficient in both CD80 and CD86 developed less atherosclerosis and the effector T cells in their spleen and lymph nodes produced less IFN-γ, suggesting that the CD28-CD80/86 dyad regulates atherosclerosis by priming T cells.[55] Accordingly, mice overexpressing CTLA4 in T cells developed less atherosclerosis and a dampened T effector cell response.[56] The first results targeting the CD28-CD80/CD86 axis pharmacologically are promising. Inhibition of CD28-CD80/CD86 interactions by the CTLA4 immunoglobulin fusion protein abatacept reduced atherosclerosis in $ApoE^{-/-}$ mice.[57] Moreover, *ex vivo* blocking of CD80 in human carotid endarterectomy specimens reduced cytokine secretion.[58] CD80 has been identified as an imaging marker to detect vulnerable plaques, both *ex vivo* in human atherosclerotic plaques and *in vivo* in atherosclerotic arteries of $ApoE^{-/-}$ mice.[59,60] CD80-specific non-invasive imaging may thus be a promising tool to discriminate stable and vulnerable atherosclerotic plaques and the CD28-CD80/86 dyad a potential therapeutic target for plaque stabilization.

Another potent immune-checkpoint target for CVD is the CD40L-CD40 dyad. CD40-CD40L interaction not only enhances T-cell stimulation and mediates T cell help for cytotoxic T-cell priming, but also induces macrophage, DC and B-cell activation and proliferation and is crucial for Ig-isotype switching.[61] Genetic deletion or antibody-mediated inhibition of CD40L or CD40 in $ApoE^{-/-}$ mice and $LDLR^{-/-}$ mice decreased atherosclerotic plaque burden and induced collagen-rich plaques containing few immune cells, i.e. the murine equivalent of a stable plaque.[62–66] In humans, the expression of the CD40L-CD40 dyad in the plaque was associated with histological features of plaque vulnerability. The soluble form of CD40L, sCD40L is associated with hypercholesterolaemia, stroke, diabetes, and acute coronary syndrome and can predict recurrent CVD.[67]

CD40 signals via TNF-receptor-associated factors (TRAFs), and the interaction between CD40-TRAF6, but not CD40-TRAF2/3/5, was found to be crucial in atherogenesis.[66] Mice lacking CD40-TRAF6 interactions exhibited impaired arterial leucocyte recruitment and low plaque macrophage content. The majority of $CD40\text{-}TRAF6^{-/-}$ monocytes in blood and spleen is of the patrolling $Ly6C^{low}$ subtype, and $CD40\text{-}TRAF6^{-/-}$ macrophages were anti-inflammatory.[66]

One of the risks of blocking CD40-CD40L interactions during a long-time treatment is immune suppression. To circumvent this problem, small molecule inhibitors (SMIs) that selectively block the interaction between CD40 and TRAF6 (TRAF-STOPs), without affecting CD40-TRAF2/3/5 interactions, thereby keeping CD40-mediated immunity such as Ig-isotype switching and T-cell proliferation intact, have been developed.[68–70] In a diet-induced obesity model, blocking CD40-TRAF6 interactions by SMI 6860766 or 6877002 reduced adipose tissue inflammation, haepatosteatosis, and improved insulin sensitivity.[69,71] In other models, SMI 6877002 decreased neuroinflammation and prevented allograft reaction when given as combination therapy with rapamycin in a nanomedicinal setting, and reduces adverse cardiac remodelling after pressure overload induced HF.[72–74] TRAF-STOP treatment with either SMI 6860766 or 6877002 reduced early atherosclerosis development and halted the progression of established atherosclerosis in $ApoE^{-/-}$ mice.[70] Moreover, TRAF-STOP treatment prevented classical monocyte activation, leucocyte recruitment, macrophage activation, and migration in the arterial wall and induced a stable plaque phenotype.[70] Nanomedicinal delivery of TRAF-STOPs specifically to macrophages, by incorporating SMI 6877002 into recombinant high-density lipoprotein (rHDL) nanoparticles, reduced atherosclerotic plaque formation when applied as prevention therapy, and attenuated macrophage accumulation in established plaques.[70,75] Moreover, nanotherapeutic delivery of rHDL-6877002 was safe, and had a favourable biodistribution in non-human primates.[75]

Immune checkpoint therapy is a powerful therapeutic strategy to treat atherosclerosis. However, off-target effects might encompass immune suppression. Highly selective immune checkpoint modulation by intervening in cell-specific downstream signalling pathways, or targeted drug delivery will render immune checkpoint-therapies for CVD feasible and safe.

Metabolic regulation of immune cells in atherosclerosis

Activation of immune cells goes hand in hand with alterations in intracellular metabolic pathways, and metabolites are in fact master switches of immune-cell functions, making them excellent targets to counteract inflammation.[76] Immune cells use several metabolic traits to adapt to the wide array of metabolic demands seen during inflammation in CVD, specifically glycolysis, the tricarboxylic acid (TCA) cycle, the pentose phosphate pathway (PPP), mitochondrial fatty acid β-oxidation and oxidative phosphorylation (OXPHOS), fatty acid synthesis, and metabolism of amino acids (AAs).[76] Metabolic profiling of human atherosclerotic plaques showed that symptomatic CVD is characterized by increased inflammation and histologically high vulnerability along with decreased fatty acid oxidation (FAO) and increased AA utilization as well as increased glycolysis. Hence, metabolic pathways seem promising targets for diagnostic purposes, e.g. using metabolic radiotracers as well as for therapeutic interventions in CVD (*Figure 2A*).[77]

Various cell types are responsible for the persistent inflammation during plaque progression including endothelial cells, innate and adaptive immune cells, all of which show distinct metabolic signatures (*Figure 2A*). For example, activated macrophages are characterized by enhanced glycolysis to enable sufficient ATP production to exert their effector function including phagocytosis and cytokine production. In CVD, overutilization of glucose promotes excessive and prolonged production of the cytokines IL-6 and IL-1β, presumably through the glycolytic enzyme pyruvate kinase M2 (PKM2)/signal

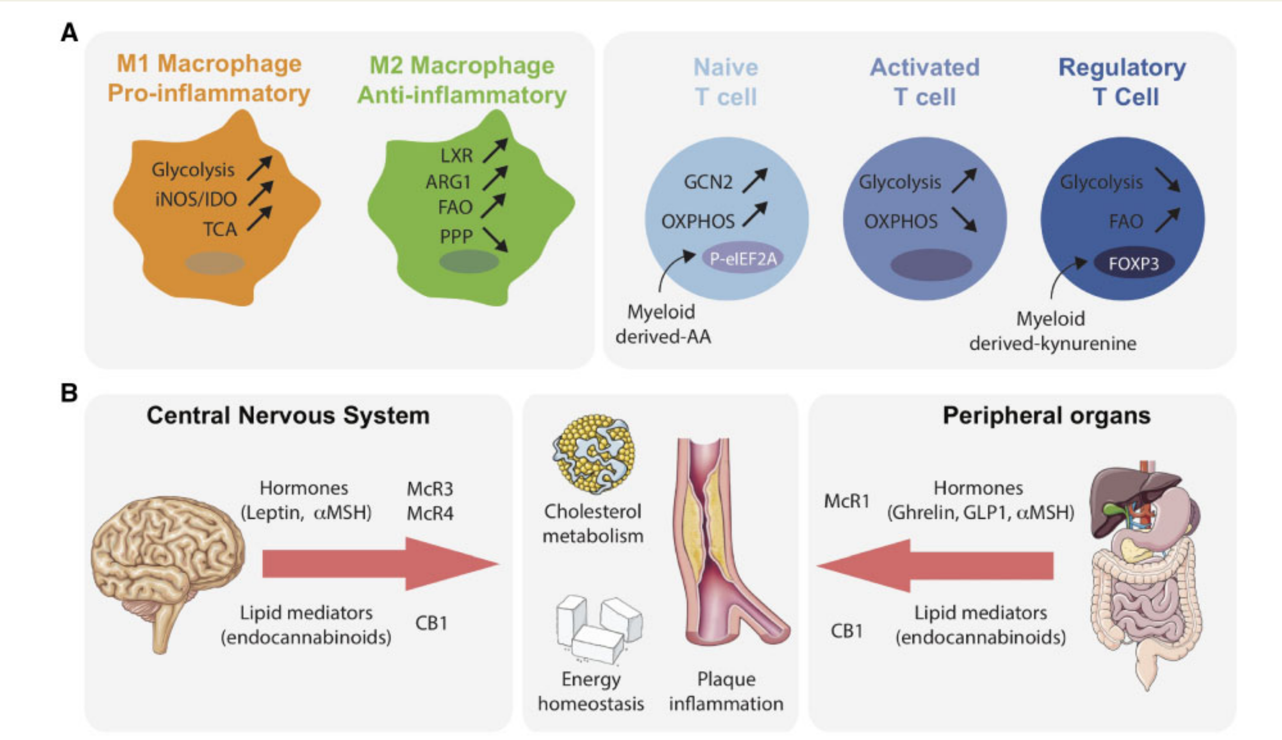


Figure 2 Implication of immunometabolism, hormones, and lipid mediators in cardiovascular disease. (*A*) Metabolic regulation of macrophages and T cells in cardiovascular disease. Pro-inflammatory M1 type macrophages show enhanced glycolysis and tricarboxylic acid cycle (TCA) activity as well as induced expression of the amino acid (AA) degrading enzymes indolamine 2,3-dioxygenase 1 (IDO1) and inducible nitric oxide synthase (iNOS). Anti-inflammatory M2 type macrophages are characterized by increased liver X receptor (LXR) activation and arginase 1 (ARG1) expression and by enhanced fatty acid β-oxidation (FAO) and suppressed activity of the pentose phosphate pathway (PPP). Myeloid-mediated AA starvation activates T-cell stress kinase GCN2 to phosphorylate eukaryotic initiation factor2a (P-elF2a). Upon activation, naïve T cells shift from oxidative phosphorylation (OXPHOS) towards glycolysis. Inhibition of glycolysis directs T cells into an immuno-regulatory Treg phenotype. The IDO1-metabolite kynurenine also induce expression of Forkhead box P3 (FoxP3), the lineage-defining transcription factor for Tregs. (*B*) Endocrine regulation of cholesterol metabolism, plaque inflammation and energy homeostasis in cardiovascular disease. Hypothalamic regulation of energy balance and liver lipid metabolism involves melanocortin receptor signalling (McR3, McR4 and the agonist alpha melanocyte-stimulating hormone, αMSH), which is influenced by gut hormones ghrelin and glucagon like peptide 1 (GLP1). In addition, MCR1 expressed by plaque macrophages represents a counter-regulatory mechanism against plaque cholesterol accumulation and inflammation in atherosclerosis. Brain CB1 receptors are involved in central regulation of energy homeostasis. There is emerging evidence for peripheral CB1 signalling controlling adipose tissue, liver, and pancreas metabolic functions as well as inflammation. A pro-atherogenic role for CB1 in macrophage cholesterol metabolism was shown *in vitro*; however, the cell-specific function of CB1 in atherosclerosis deserves further clarification *in vivo*.

transducer and activator (STAT)3 pathway.[78] Furthermore, hexokinase 1, another glycolytic enzyme was identified to activate the NLRP3 inflammasome,[79] the main pathway for IL-1β production. Besides a pro-inflammatory plaque macrophage population, plaques contain lipid-loaded macrophages that are more anti-inflammatory in nature.[80] These foamy macrophages are characterized by increased liver X receptor (LXR) activation, and suppressed activity of the PPP in foam cells, which causes the reduced inflammatory responses in these cells.[81]

Distinct AA, such as L-arginine or L-tryptophan, and their metabolism via arginase 1 (ARG1), inducible nitric oxide synthase (iNOS) and indolamine 2,3-dioxygenase 1 (IDO1) substantially control macrophage polarization[82] and proliferation of surrounding adaptive immune cells, due to starvation of their local AA environment. Experimental evidence in atherosclerosis is still limited although recent reports revealed that haematopoietic ARG1 prevented foam cell formation[83] and IDO1 deficiency reduced plaque size.[84] However, definitive roles of ARG1 and IDO1 in atherosclerosis remain controversial and warrant further studies.

In addition to glucose and AA metabolism, M1 macrophages are hallmarked by alterations in the TCA trait leading to accumulation of pro-inflammatory TCA-metabolites, such as citrate[85] and succinate,[86] while M2 macrophages show a rather intact TCA cycle. Besides disposing the TCA cycle, M2 macrophages also rely on STAT6/peroxisome proliferator activated receptor gamma co-activator (PGC)1ß-mediated FAO which was found to induce anti-inflammatory signalling.[87] Thus, metabolic priming by therapeutics that promote utilization of fatty acids, such as fibric acids and thiazolidinediones, may be a promising approach to attenuate macrophage-mediated inflammation in atherosclerosis.

Cells of the adaptive immune system also drive their activation and polarization via changes in metabolism. Upon activation, naïve T cells shift towards glycolysis to ensure fast energy supply.[88,89] Inhibition of glycolysis effectively shifts T cells from a Th17 into an immune-regulatory Treg phenotype.[90] To regulate effector and suppressor T-cell homeostasis, T cells constitute FAO, with anti-inflammatory Tregs showing increased FAO, as compared with T helper cells.[88] Conversely, FAO was found to promote the generation and maintenance of long-lived memory CD8+ T cells.[91] Amino acids crucially contribute to immune signalling in T cells. For example, arginine and tryptophan depletion by myeloid cells is sensed by the T-cell stress kinase GCN2-eIF2α axis, which ultimately leads to the arrest of T-cell proliferation.[92] Furthermore, myeloid-derived kynurenine, the main IDO-mediated tryptophan metabolite showed promotion of Treg differentiation and thereby limited inflammation and atherosclerosis.[93] Thus, targeting arginine and tryptophan metabolism as well as their downstream metabolites may provide interesting strategies for clinical interventions.

Taken together, various metabolic pathways in innate and adaptive immune cells open an intriguing avenue for new anti-inflammatory therapies for CVD targeting specific aspects of the immune system.

Peripheral and neuronal control of cardiovascular disease via hormones and lipid mediators

Besides classical immunomodulators, novel evidence has revealed that the endocrine system is a strong mediator of both inflammation and lipid metabolism in CVD. The identification of neuroendocrine changes in the hypothalamus that regulate cholesterol metabolism in the liver was a groundbreaking discovery that has opened a completely new avenue in searching novel therapeutic possibilities for CVD in this area (*Figure 2B*).[94] Of particular interest is the development of novel targeted poly-agonists integrating gut hormones and steroids into hybrid molecules for targeting hypothalamic receptors.[95–97] In addition to these central pathways, there is emerging evidence that related peripheral hormone and lipid pathways regulate not only macrophage cholesterol metabolism, but also vascular inflammation and plaque stability.[98–106]

At the central level, melanocortin receptor (McR) signalling is crucially involved in these neuro-endocrine hypothalamic control mechanisms and is influenced by gut hormones such as ghrelin and glucagon like peptide 1.[94] A key neuroendocrine hormone in the regulation of energy balance is the adipokine leptin, which acts on the central nervous system (CNS) to regulate feeding, energy, and lipid metabolism.[107] These effects are partially mediated by the leptin interaction with proopiomelanocortin expressing neurons.[108] Proopiomelanocortin is a pro-hormone and its cleavage produces various signalling peptides, including the McR agonist alpha melanocyte-stimulating hormone (αMSH). There are two McRs expressed in the CNS, McR3 and McR4, which are both critical for the control of energy balance and liver lipid metabolism.[94,109] Despite growing evidence for these neuro-endocrine pathways in modulating lipoprotein metabolism, many mechanistic components of the CNS-mediated control in lipoprotein metabolism, inflammation, and possibly atherosclerotic plaque development remain to be determined. In particular, other relevant neuro-endocrine pathways contributing to lipoprotein metabolism and atherosclerotic plaque development might involve hypothalamic CB1 cannabinoid receptors. CB1 is highly expressed in the hypothalamus, and its blockade improves atherogenic dyslipidaemia.[110] Moreover, the role of hypothalamic inflammation in the pathogenesis of atherosclerosis, beyond its documented role in lipoprotein and energy metabolism,[111,112] as well as the role of hypothalamic circuits involving hypocretin in the modulation of haematopoiesis and myeloid-driven atherogenesis by sleep,[111,112] deserves further investigation.

The endocannabinoid system is an endogenous lipid signalling system that regulates several pathways in the CNS and peripheral tissues and is reported to modulate inflammation.[113] It comprises a group of arachidonic acid-derived endogenous lipid mediators (endocannabinoids), which bind to cannabinoid receptors CB1 and CB2. An over-activity of the endocannabinoid system plays a pathophysiological role in the development of metabolic disorders and atherosclerosis.[101] Recent evidence suggests that enhanced endocannabinoid signalling exerts multiple effects in atherosclerosis, including a modulation of vascular inflammation, leucocyte recruitment, macrophage cholesterol metabolism, and consequently plaque stability.[98–102] In addition, recent findings in various metabolic disease models highlight the relevance of peripheral CB1 cannabinoid receptors in adipose tissue, liver and pancreas, which crucially regulate lipid and glucose metabolism as well as macrophage properties in these organs.[113–116] This suggests that blocking CB1 signalling in the vasculature and peripheral organs might have a potent therapeutic benefit for atherosclerosis by inhibiting vascular inflammation and improving metabolic risk factors.

A novel link has been established between peripheral McR signalling and macrophage cholesterol metabolism.[104] McR1 is abundantly expressed by melanocytes in the skin where it stimulates melanin production. The same receptor is also present in macrophages in human and mouse atherosclerotic plaques.[104] Notably, the amount of McR1 in plaques correlated with transporters that are known to unload excess cholesterol from macrophages, namely ABCA1 and ABCG1. This suggests that McR1 could serve as a protective regulator of macrophage cholesterol transport. In support of this hypothesis, *in vitro* activation of McR1 in primary murine macrophages reduced the uptake of oxidized LDL and enhanced the removal of intracellular cholesterol through its efflux.[104] Conversely, McR1-deficient macrophages had impaired cholesterol efflux and increased tendency to foam cell formation. A subsequent study confirmed these findings *in vivo*. Deficiency in Mc1R1 signalling in ApoE$^{-/-}$ mice was associated with significantly larger atherosclerotic lesions upon Western diet, reduced aortic ABCA1 and ABCG1 expression as well as enhanced hypercholesterolaemia and hepatic lipid accumulation.[98] These findings may have therapeutic implications in atherosclerosis, since macrophage cholesterol efflux is an important counter-regulatory mechanism against cholesterol accumulation and inflammation in atherosclerotic plaques. Although the field of (neuro)-endocrine regulation of CVD is still emerging, modulating (neuro)endocrine signalling was found to have profound effects in the vasculature and metabolically relevant organs and might therefore have a potent therapeutic benefit for atherosclerosis by inhibiting vascular inflammation and improving metabolic risk factors.

Immunotherapy for cardiovascular disease: Per aspera ad astra?

The CANTOS trial has provided intriguing proof-of-principle for the inflammatory hypothesis of atherosclerosis in man and has paved the way for the feasibility of immunotherapy for CVD. Although canakinumab treatment lowered the risk for primary (MI, stroke, CV death) and secondary endpoints significantly, CANTOS did not show a decrease in overall and CV mortality. Moreover, side effects of canakinumab treatment included leucopenia and a higher rate of fatal infection.[11] These data warrant further studies to identify patient subgroups in which atherosclerosis is particularly driven by inflammation, and who would benefit most from anti-inflammatory treatment.[17] Alternative immunotherapeutic targets, as outlined in this review (*Take home figure*), have a high potential to be used as immunotherapy for atherosclerosis and should be pursuit. Given their advanced specificity, some of them may have the potential to reduce CVD and CV mortality to a greater extent than canakinumab. Moreover, strategies, such as targeting chemokine heterodimers, creating polyagonists/antagonists for hormones or intervening in specific downstream signalling pathways of immune checkpoints, may strongly reduce atherosclerosis and CVD, while causing less immunosuppressive side-effects. Most recently, complex formation of ApoE with activated complement C1q attenuating the classical complement cascade has been identified as a hallmark for persistent inflammation in choroid plexus and atherosclerotic plaques of mice and humans. Whereas deficiency in ApoE exacerbated complement and disease activity, silencing of the complement precursor C5 reduced inflammation and disease burden in mouse models of both Alzheimer's disease and atherosclerosis.[124] Advances in immunology have enabled and accelerated our understanding of the complex immune network involved in the initiation and progression of atherosclerosis. Since CANTOS has epitomized the importance of targeting inflammation in atherosclerosis, it is now time to perfect immunotherapy for CVD, which will be a difficult yet worthwhile path.

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